

Problem Statement

Business Context

The healthcare industry is rapidly evolving, with professionals facing increasing challenges in managing vast volumes of medical data while delivering accurate and timely diagnoses. The need for quick access to comprehensive, reliable, and up-to-date medical knowledge is critical for improving patient outcomes and ensuring informed decision-making in a fast-paced environment.

Healthcare professionals often encounter information overload, struggling to sift through extensive research and data to create accurate diagnoses and treatment plans. This challenge is amplified by the need for efficiency, particularly in emergencies, where time-sensitive decisions are vital. Furthermore, access to trusted, current medical information from renowned manuals and research papers is essential for maintaining high standards of care.

To address these challenges, healthcare centers can focus on integrating systems that streamline access to medical knowledge, provide tools to support quick decision-making, and enhance efficiency. Leveraging centralized knowledge platforms and ensuring healthcare providers have continuous access to reliable resources can significantly improve patient care and operational effectiveness.

Common Questions to Answer

- 1. Diagnostic Assistance:** "What are the common symptoms and treatments for pulmonary embolism?"
- 2. Drug Information:** "Can you provide the trade names of medications used for treating hypertension?"
- 3. Treatment Plans:** "What are the first-line options and alternatives for managing rheumatoid arthritis?"
- 4. Specialty Knowledge:** "What are the diagnostic steps for suspected endocrine disorders?"
- 5. Critical Care Protocols:** "What is the protocol for managing sepsis in a critical care unit?"

Objective

As an AI specialist, your task is to develop a RAG-based AI solution using renowned medical manuals to address healthcare challenges. The objective is to **understand** issues like information overload, **apply** AI techniques to streamline decision-making, **analyze** its impact on diagnostics and patient outcomes, **evaluate** its potential to standardize care practices, and **create** a functional prototype demonstrating its feasibility and effectiveness.

Data Description

The **Merck Manuals** are medical references published by the American pharmaceutical company Merck & Co., that cover a wide range of medical topics, including disorders, tests, diagnoses, and

drugs. The manuals have been published since 1899, when Merck & Co. was still a subsidiary of the German company Merck.

The manual is provided as a PDF with over 4,000 pages divided into 23 sections.

Installing and Importing Necessary Libraries and Dependencies

```
# Installation for GPU llama-cpp-python
# uncomment and run the following code in case GPU is being used
#!CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 pip install llama-cpp-
python==0.2.28 --force-reinstall --no-cache-dir -q

# Installation for CPU llama-cpp-python
# uncomment and run the following code in case GPU is not being used
#!CMAKE_ARGS="-DLLAMA_CUBLAS=off" FORCE_CMAKE=1 pip install llama-cpp-
python==0.2.28 --force-reinstall --no-cache-dir -q

!CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 \
pip install llama-cpp-python==0.2.28 --force-reinstall --no-cache-dir

Collecting llama-cpp-python==0.2.28
  Downloading llama_cpp_python-0.2.28.tar.gz (9.4 MB)
    _____ 0.0/9.4 MB ? eta -:-:-:-
    _____ 1.4/9.4 MB 42.9 MB/s eta
0:00:01 _____ 9.4/9.4 MB 147.1 MB/s
eta 0:00:00
ents to build wheel ... etadata (pyproject.toml) ... llama-cpp-
python==0.2.28)
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3
kB)
Collecting numpy>=1.20.0 (from llama-cpp-python==0.2.28)
  Downloading numpy-2.4.3-cp312-cp312-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Collecting diskcache>=5.6.1 (from llama-cpp-python==0.2.28)
  Downloading diskcache-5.6.3-py3-none-any.whl.metadata (20 kB)
  Downloading diskcache-5.6.3-py3-none-any.whl (45 kB)
    _____ 45.5/45.5 kB 301.7 MB/s eta
0:00:00
py-2.4.3-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
(16.6 MB)
    _____ 16.6/16.6 MB 340.1 MB/s eta
0:00:00
    _____ 44.6/44.6 kB 294.4 MB/s eta
0:00:00
a-cpp-python
  Building wheel for llama-cpp-python (pyproject.toml) ... a-cpp-
python: filename=llama_cpp_python-0.2.28-cp312-cp312-linux_x86_64.whl
size=8960012
sha256=f5dc615002485eab0b0fecda14fa7fadc9deb360622585a904a5064d1300b62
```

4

```
Stored in directory:
/tmp/pip-ephem-wheel-cache-6tnlvu2y/wheels/f6/3d/56/3f3ff71bc61e1aa756
c6d6fcd36905032bc5868d3a5bd39d60
Successfully built llama-cpp-python
Installing collected packages: typing-extensions, numpy, diskcache,
llama-cpp-python
  Attempting uninstall: typing-extensions
    Found existing installation: typing_extensions 4.15.0
    Uninstalling typing_extensions-4.15.0:
      Successfully uninstalled typing_extensions-4.15.0
  Attempting uninstall: numpy
    Found existing installation: numpy 2.3.3
    Uninstalling numpy-2.3.3:
      Successfully uninstalled numpy-2.3.3
  Attempting uninstall: diskcache
    Found existing installation: diskcache 5.6.3
    Uninstalling diskcache-5.6.3:
      Successfully uninstalled diskcache-5.6.3
  Attempting uninstall: llama-cpp-python
    Found existing installation: llama_cpp_python 0.2.28
    Uninstalling llama_cpp_python-0.2.28:
      Successfully uninstalled llama_cpp_python-0.2.28
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
langgraph-prebuilt 1.0.8 requires langchain-core>=1.0.0, but you have
langchain-core 0.3.83 which is incompatible.
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy
2.4.3 which is incompatible.
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-
sdk<1.39.0,>=1.35.0, but you have opentelemetry-sdk 1.40.0 which is
incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-
exporter-otlp-proto-common==1.38.0, but you have opentelemetry-
exporter-otlp-proto-common 1.40.0 which is incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-
proto==1.38.0, but you have opentelemetry-proto 1.40.0 which is
incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-
sdk~=1.38.0, but you have opentelemetry-sdk 1.40.0 which is
incompatible.
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.4.3 which
is incompatible.
google-adk 1.27.1 requires opentelemetry-api<1.39.0,>=1.36.0, but you
have opentelemetry-api 1.40.0 which is incompatible.
google-adk 1.27.1 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you
have opentelemetry-sdk 1.40.0 which is incompatible.
```

Successfully installed diskcache-5.6.3 llama-cpp-python-0.2.28 numpy-2.4.3 typing-extensions-4.15.0

Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in *this notebook*.

```
# For installing the libraries & downloading models from HF Hub
!pip install huggingface_hub==0.35.3 pandas==2.2.2 tiktoken==0.12.0
pymupdf==1.26.5 langchain==0.3.27 langchain-community==0.3.31
chromadb==1.1.1 sentence-transformers==5.1.1 numpy==2.3.3 -q
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy 2.3.3 which is incompatible.
numba 0.60.0 requires numpy<2.1,>=1.22, but you have numpy 2.3.3 which is incompatible.

Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in *this notebook*.

```
#Libraries for processing dataframes,text
```

```
import json,os
import tiktoken
import pandas as pd
```

```
#Libraries for Loading Data, Chunking, Embedding, and Vector Databases
```

```
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings.sentence_transformer import
SentenceTransformerEmbeddings
from langchain_community.vectorstores import Chroma
```

```
#Libraries for downloading and loading the llm
```

```
from huggingface_hub import hf_hub_download
from llama_cpp import Llama
```

Question Answering using LLM

Downloading and Loading the model

```
model_name_or_path = "TheBloke/Mistral-7B-Instruct-v0.2-GGUF"
model_basename = "mistral-7b-instruct-v0.2.Q6_K.gguf"

model_path = hf_hub_download(
    repo_id=model_name_or_path,
    filename=model_basename
)

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
    warnings.warn(

llm = Llama(
    model_path=model_path,
    n_ctx=2048,
    n_gpu_layers=-1,
    n_batch=256,
    verbose=True
)

AVX = 1 | AVX_VNNI = 0 | AVX2 = 1 | AVX512 = 1 | AVX512_VBMI = 0 |
AVX512_VNNI = 0 | FMA = 1 | NEON = 0 | ARM_FMA = 0 | F16C = 1 |
FP16_VA = 0 | WASM_SIMD = 0 | BLAS = 1 | SSE3 = 1 | SSSE3 = 1 | VSX =
0 |
```

Response

```
def response(query, max_tokens=512, temperature=0, top_p=0.95,
top_k=50):
    prompt = f"<s>[INST] {query} [/INST]"

    output = llm(
        prompt=prompt,
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p,
        top_k=top_k
    )

    return output["choices"][0]["text"].strip()
```

```
print(response("What treatment options are available for managing low blood sugar?"));
```

Low blood sugar, also known as hypoglycemia, can be caused by various conditions including diabetes, certain medications, or other health issues. Here are some common treatment options for managing low blood sugar:

1. Consuming carbohydrates: The quickest way to raise your blood sugar level is by consuming foods that contain carbohydrates. These include fruits, juices, sweets, and starches like bread, rice, or pasta. Aim for 15-20 grams of carbohydrates, which is roughly equivalent to a medium apple, a cup of milk, or 3-4 glucose tablets.
2. Glucose tablets or gel: These are readily available over the counter and can be carried with you for quick treatment of low blood sugar. They provide a fast source of glucose to raise your blood sugar level.
3. Dextrose solution: Injectable dextrose solutions, such as D50W (50% dextrose in water), are prescribed by doctors for severe cases of hypoglycemia or when oral treatments are not effective. These should only be used under medical supervision.
4. Medications: Certain medications like glucagon can be injected to raise blood sugar levels in case of severe hypoglycemia. Glucagon is a hormone that stimulates the liver to release stored glucose into the bloodstream.
5. Adjusting diabetes medication: If you have diabetes, your healthcare provider may recommend adjusting your insulin or oral diabetes medications to prevent low blood sugar episodes. This might involve changing the dosage, timing of administration, or type of medication.
6. Regular monitoring: Monitoring your blood sugar levels regularly can help you identify patterns and triggers for low blood sugar episodes. Keep a log of your readings and discuss any concerns with your healthcare provider to develop an effective management plan.
7. Lifestyle modifications: Making lifestyle changes, such as eating balanced meals, avoiding long periods between meals, and getting regular physical activity, can help prevent low blood sugar episodes in people with diabetes or other conditions that put them at risk.

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
user_query_1 = "What is the protocol for managing sepsis in a critical care unit?";
```

```
user_response_1 = response(user_query_1)
print(user_response_1)
```

Llama.generate: prefix-match hit

Sepsis is a life-threatening condition that requires prompt recognition and aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care unit:

1. Early recognition: Recognize the signs and symptoms of sepsis early, which may include fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis.
2. Quick initiation of antibiotics: Start appropriate broad-spectrum antibiotics as soon as possible based on the patient's clinical presentation and culture results if available.
3. Fluid resuscitation: Aggressively administer intravenous fluids to maintain adequate tissue perfusion, with a goal of maintaining a mean arterial pressure (MAP) of 65 mmHg or higher and a central venous oxygen saturation (ScvO2) of greater than 70%.
4. Vasopressor support: If the patient is hypotensive despite adequate fluid resuscitation, administer vasopressors to maintain MAP.
5. Corticosteroids: Consider using corticosteroids in patients with septic shock who have not responded to initial fluid resuscitation and vasopressor therapy.
6. Inotropic support: Use inotropes to improve cardiac output if necessary.
7. Mechanical ventilation: Provide mechanical ventilation if the patient is experiencing respiratory failure.
8. Source control: Identify and address the source of infection, such as removing an infected catheter or draining an abscess.
9. Monitoring: Close monitoring of vital signs, laboratory values, and hemodynamic status is essential.
10. Goal-directed therapy: Consider using goal-directed therapy to optimize fluid resuscitation and improve outcomes.
11. Early goal-directed therapy (EGDT): EGDT involves using specific targets for mean arterial pressure, central venous oxygen saturation, and urine output to guide resuscitation efforts.
12. Septic shock protocols: Follow a septic shock protocol or bundle of care, which may include early recognition, rapid initiation of antibiotics, fluid resuscitation, vasopressor therapy, and source control.
13. Continuous renal replacement therapy (CRRT)

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_query_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?";  
user_response_2 = response(user_query_2)  
print(user_response_2)
```

Llama.generate: prefix-match hit

Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The following are common symptoms of appendicitis:

1. Abdominal pain: The pain typically starts as a mild discomfort around the navel or upper abdomen that may come and go. However, it can quickly develop into a sharp, persistent pain in the lower right side of the abdomen.
2. Loss of appetite: People with appendicitis often lose their appetite due to abdominal pain and discomfort.
3. Nausea and vomiting: Vomiting is common in appendicitis, especially as the condition progresses.
4. Fever: A fever of 100.4°F (38°C) or higher is a common symptom of appendicitis.
5. Constipation or diarrhea: Some people with appendicitis experience constipation, while others have diarrhea.
6. Abdominal swelling: The abdomen may become swollen and tender to the touch.
7. Inability to pass gas: Passing gas can be difficult due to the pressure on the intestines caused by appendicitis.

Appendicitis cannot be cured with medicine alone, as the inflammation and infection in the appendix require surgical intervention. The standard treatment for appendicitis is an appendectomy, which involves removing the appendix through a small incision in the abdomen or through keyhole surgery (laparoscopy). This procedure is typically performed under general anesthesia and can be done as an open or laparoscopic procedure. The goal of the surgery is to remove the inflamed appendix before it ruptures, which could lead to a more serious infection in the abdominal cavity (peritonitis). After surgery, patients usually recover within a few days and can resume their normal activities within a week or two.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_query_3 = "What are the effective treatments or solutions for  
addressing sudden patchy hair loss, commonly seen as localized bald  
spots on the scalp, and what could be the possible causes behind it?";  
user_response_3 = response(user_query_3)  
print(user_response_3)
```

Llama.generate: prefix-match hit

Sudden patchy hair loss, also known as alopecia areata, is an autoimmune condition that causes hair loss in small patches on the scalp or other areas of the body. The exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system. Here are some effective treatments and possible causes for this condition:

1. Corticosteroids: Corticosteroids are the most common treatment for alopecia areata. They work by reducing inflammation and suppressing the immune system, which helps to promote hair regrowth. Corticosteroids can be applied topically or taken orally in the form of pills or injections.
2. Immunotherapy: Immunotherapies such as diphenylcyclopropenone (DPCP) and squaric acid dibutylester (SADBE) are used to stimulate an immune response that promotes hair regrowth. These treatments can be applied topically or given as injections.
3. Minoxidil: Minoxidil is a medication that is commonly used to treat male and female pattern baldness, but it may also be effective for alopecia areata. It works by stimulating hair growth and prolonging the anagen phase of the hair growth cycle.
4. Anthralin: Anthralin is a topical medication that is used to treat psoriasis, but it has also been found to be effective in treating alopecia areata. It works by reducing inflammation and suppressing the immune system.
5. Nutritional supplements: Certain nutritional deficiencies, such as iron deficiency or vitamin D deficiency, can contribute to hair loss. Supplementing with these nutrients may help promote hair growth in people with alopecia areata.
6. Stress reduction: Stress is a known trigger for alopecia areata. Reducing stress through techniques such as meditation, yoga, or other relaxation methods may help prevent or reduce the severity of hair loss.
7. Avoid tight hairstyles: Tight hairstyles, such as braids or ponytails, can pull on the hair and cause damage, which can contribute to hair loss. Avoiding these styles and using gentle hair care practices can help prevent further hair loss.
8. Ident

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_query_4 = "What treatments are recommended for a person who has  
sustained a physical injury to brain tissue, resulting in temporary or  
permanent impairment of brain function?";  
user_response_4 = response(user_query_4)  
print(user_response_4)
```

Llama.generate: prefix-match hit

Treatment for a physical injury to the brain can depend on the severity and location of the injury. Here are some common treatments that may be recommended:

1. Emergency care: For severe brain injuries, immediate medical attention is necessary. This may include surgery to remove hematomas or other obstructions, as well as measures to prevent further damage, such as controlling blood pressure and maintaining adequate oxygen levels.
2. Rehabilitation: Rehabilitation is an essential part of treatment for brain injuries. It can help improve function, reduce symptoms, and promote independence. Rehabilitation may include physical therapy, occupational therapy, speech therapy, and cognitive rehabilitation.
3. Medications: Depending on the specific symptoms of the brain injury, medications may be prescribed to manage conditions such as seizures, pain, or depression.
4. Assistive devices: Devices such as wheelchairs, walkers, or communication aids can help individuals with brain injuries regain mobility and independence.
5. Lifestyle modifications: Making lifestyle modifications, such as getting enough rest, eating a healthy diet, and avoiding alcohol and drugs, can help improve overall health and support recovery from a brain injury.
6. Support groups: Joining a support group can provide emotional and social support for individuals with brain injuries and their families. These groups can offer valuable information, resources, and connections to others who are going through similar experiences.
7. Alternative therapies: Some alternative therapies, such as acupuncture, massage therapy, or biofeedback, may be helpful in managing symptoms of a brain injury. However, it's important to discuss these options with a healthcare provider before trying them.
8. Vocational rehabilitation: For individuals who have sustained a brain injury that affects their ability to work, vocational rehabilitation can help them develop skills and find employment opportunities that are suitable for their new abilities.
9. Psychological support: Brain injuries can cause emotional and psychological challenges, such as depression, anxiety, or difficulty with memory and attention. Psychological support, such as counseling

or therapy, can help individuals cope with these issues and improve overall well-being.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_query_5 = "What are the necessary precautions and treatment steps  
for a person who has fractured their leg during a hiking trip, and  
what should be considered for their care and recovery?";  
user_response_5 = response(user_query_5)  
print(user_response_5)
```

Llama.generate: prefix-match hit

If a person has fractured their leg while hiking, it is important to follow these necessary precautions and treatment steps:

1. Assess the situation: Check the severity of the injury and assess if the fracture is open or closed. An open fracture is when the bone breaks through the skin, which increases the risk of infection. If the fracture is open, do not attempt to move the person as it may cause further damage or infection.
2. Call for help: If possible, call for emergency medical assistance. If there is no cell phone reception, use a satellite phone or hike back to civilization to seek help.
3. Provide first aid: Keep the person calm and comfortable. Apply a sterile dressing to any open wounds and apply ice to reduce swelling. Do not try to realign the bone as it may cause further damage. Splint the leg if possible using a makeshift splint or a commercial one if available.
4. Transportation: If the person is stable, transport them to the nearest medical facility as soon as possible. If transportation is difficult, make sure they are as comfortable as possible and keep them warm.
5. Hospital treatment: Once at the hospital, the doctor will assess the fracture and determine the best course of treatment. This may include setting the bone with pins or a cast, surgery, or other treatments depending on the severity of the injury.
6. Recovery: The recovery process for a leg fracture can take several weeks to months, depending on the severity of the injury and the individual's overall health. During this time, it is important to follow the doctor's instructions for care and rehabilitation. This may include physical therapy, medication, and avoiding weight-bearing activities.
7. Prevention: To prevent future leg fractures while hiking, make sure to wear appropriate footwear with good traction, stay on marked trails, and avoid hiking in extreme weather conditions or when

fatigued. Additionally, carrying a first aid kit and knowing basic first aid techniques can help in case of an emergency.

Question Answering using LLM with Prompt Engineering

```
prompt_template = ""
```

```
You are a medical AI assistant. Provide accurate, evidence-based, and clinically sound answers grounded in established medical guidelines and best practices.
```

```
Guidelines:
```

- Prioritize factual accuracy and medical consensus.
- Use clear, concise language appropriate for both healthcare professionals and general audiences.
- Structure responses logically (e.g., definition, causes, symptoms, diagnosis, treatment when relevant).
- Avoid unnecessary verbosity or speculation.
- If information is uncertain, incomplete, or outside your knowledge scope, explicitly acknowledge the limitation.

```
Safety:
```

- Do not provide diagnoses or treatment plans as a substitute for professional medical care.
- Encourage seeking qualified healthcare professionals when appropriate.

```
Answer the question directly and clearly.
```

```
"""
```

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
query_1 = prompt_template + "\n" + "What is the protocol for managing sepsis in a critical care unit?";  
print(response(query_1))
```

```
Llama.generate: prefix-match hit
```

Sepsis is a life-threatening condition caused by the body's response to an infection, which can lead to tissue damage, organ failure, and even death if not recognized and treated promptly. In a critical care unit, managing sepsis involves a systematic approach based on established guidelines from organizations like the Surviving Sepsis Campaign (SSC). Here are the key steps:

1. **Recognition**: Suspect sepsis in any patient with suspected or confirmed infection and signs of organ dysfunction such as altered mental status, respiratory distress, lactic acidosis, or low urine output.

2. **Early goal-directed therapy (EGDT)**: Initiate resuscitation measures within 1 hour of recognition:
 - Administer oxygen to maintain adequate oxygenation and ventilate if necessary.
 - Provide intravenous fluids to restore intravascular volume, aiming for a mean arterial pressure (MAP) >65 mmHg and central venous pressure (CVP) 8-12 cm H2O or a MAP >70 mmHg and CVP <12 cm H2O if no CVP measurement is available.
 - Administer broad-spectrum antibiotics based on suspected infection site and microbiology results.
3. **Source control**: Identify and address the source of infection as soon as possible, which may involve surgical intervention or other interventions like drainage or debridement.
4. **Monitoring**: Continuously monitor vital signs, lactate levels, urine output, and organ function (e.g., renal, liver, cardiac).
5. **Sequential Organ Failure Assessment (SOFA) score**: Calculate the SOFA score to assess the severity of organ dysfunction.
6. **Corticosteroids**: Consider administering corticosteroids in cases of septic shock or refractory hypotension, but only after adequate fluid resuscitation and source control have been addressed.
7. **Vasopressors**: Use vasopressors to maintain MAP >65 mmHg if fluid resuscitation alone is insufficient.
8. **Inotropes**: Consider using inotropic agents to improve cardiac output if needed.
9. **Ins**

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
query_2 = prompt_template + "\n" + "What are the common symptoms for  
appendicitis, and can it be cured via medicine? If not, what surgical  
procedure should be followed to treat it?";  
print(response(query_2))
```

Llama.generate: prefix-match hit

Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right abdomen. Common symptoms include:

1. Abdominal pain: The pain typically starts around the navel area and

then shifts to the lower right side of the abdomen. It may be constant or intermittent and worsens with movement, coughing, or sneezing.

2. Loss of appetite: Affected individuals may experience a decrease in appetite due to abdominal discomfort.
3. Nausea and vomiting: These symptoms are common in appendicitis and can occur due to the irritation of the appendix or as a response to the body's inflammatory response.
4. Fever: A fever may be present, with temperatures ranging from 100°F (37.8°C) to 103°F (39.4°C).
5. Rebound tenderness: This is a sign of peritonitis, an inflammation of the peritoneum, which may occur when appendicitis ruptures. When pressure is applied to the abdomen during a physical examination, there is pain and discomfort in response.
6. Rigidity: The affected area of the abdomen may be rigid due to muscle guarding, which occurs as a protective mechanism against further irritation or injury.

Appendicitis cannot be cured with medicine alone. If left untreated, it can lead to complications such as perforation and peritonitis, which require surgical intervention. The standard treatment for appendicitis is an appendectomy, a surgical procedure to remove the inflamed appendix. This surgery can be performed laparoscopically (using small incisions and a camera) or open (with a larger incision). Early diagnosis and treatment are crucial to prevent complications and improve outcomes. If you suspect appendicitis, consult a healthcare professional immediately.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
query_3 = prompt_template + "\n" + "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?";  
print(response(query_3))
```

Llama.generate: prefix-match hit

Sudden patchy hair loss, also known as alopecia areata, is an autoimmune condition that results in hair loss from specific areas of the scalp or other parts of the body. The exact cause of alopecia areata is not fully understood, but it's believed to be related to a problem with the immune system.

Possible causes of sudden patchy hair loss include:

1. Genetics: There seems to be a higher risk for developing alopecia areata if there is a family history of the condition.
2. Stress: Emotional or physical stress can trigger the onset of

alopecia areata in some people.

3. Viral infections: Some viruses, such as Epstein-Barr virus and human immunodeficiency virus (HIV), have been linked to alopecia areata.
4. Other autoimmune diseases: People with conditions like thyroid disease, vitiligo, or lupus may be more likely to develop alopecia areata.
5. Certain medications: Some drugs, such as certain types of antidepressants and nonsteroidal anti-inflammatory drugs (NSAIDs), have been associated with hair loss.

Effective treatments for addressing sudden patchy hair loss include:

1. Topical corticosteroids: These are applied directly to the affected area to help reduce inflammation and promote regrowth.
2. Injections of corticosteroids: In some cases, a healthcare professional may inject corticosteroids into the bald spots to stimulate hair growth.
3. Minoxidil: This medication is available over-the-counter and can be applied topically to help slow down hair loss and promote regrowth.
4. Diphencyprone (DPCP): This is a contact sensitizer that can be used to stimulate an immune response, which may help promote hair regrowth in some people with alopecia areata.
5. Systemic corticosteroids: In severe cases, oral corticosteroids may be prescribed to help reduce inflammation and promote hair growth.
6. Immunotherapy: Certain immunotherapies, such as intralesional squaric acid dibutyl ester (SADBE) or anthralin, can be used to stimulate an

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
query_4 = prompt_template + "\n" + "What treatments are recommended  
for a person who has sustained a physical injury to brain tissue,  
resulting in temporary or permanent impairment of brain function?";  
print(response(query_4))
```

Llama.generate: prefix-match hit

A brain injury, also known as traumatic brain injury (TBI), can result from various causes such as motor vehicle accidents, falls, sports injuries, or violence. Treatment for a brain injury depends on the severity and specific type of injury. Here are some common treatments:

1. Initial care: For acute brain injuries, the primary focus is on ensuring the person's safety, maintaining adequate breathing and circulation, and preventing further harm. This may involve emergency surgery to remove hematomas or decompress skull fractures, intubation for airway protection, and administration of medications to manage

increased intracranial pressure or seizures.

2. Rehabilitation: Rehabilitation is a crucial component of brain injury treatment. It can help improve functional abilities, reduce symptoms, and promote recovery. Rehabilitation may include physical therapy, occupational therapy, speech-language therapy, cognitive rehabilitation, and psychosocial support.

3. Medications: Depending on the specific symptoms and complications of a brain injury, various medications may be prescribed to manage conditions such as seizures, pain, depression, anxiety, or sleep disturbances.

4. Assistive devices: People with brain injuries may benefit from assistive devices like wheelchairs, walkers, braces, communication aids, or adaptive equipment for daily living activities.

5. Surgery: In some cases, surgery may be necessary to address underlying conditions contributing to the brain injury or its symptoms. For example, surgical intervention might be required to remove blood clots, repair skull fractures, or alleviate pressure on the brain.

6. Lifestyle modifications: Making lifestyle changes can help improve overall health and support recovery from a brain injury. This may include getting enough rest, eating a healthy diet, engaging in regular physical activity, avoiding alcohol and drugs, and practicing stress management techniques.

7. Support groups: Joining a support group can provide emotional and social benefits for individuals with brain injuries and their families. These groups offer opportunities to connect with others who have experienced similar challenges and learn from each other's experiences.

It is essential to consult healthcare professionals for personalized treatment recommendations based on the specific nature and severity of the brain injury. If you or someone you know has sustained a brain injury, seek medical attention promptly.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
query_5 = prompt_template + "\n" + "What are the necessary precautions  
and treatment steps for a person who has fractured their leg during a  
hiking trip, and what should be considered for their care and  
recovery?";  
print(response(query_5))
```


Llama.generate: prefix-match hit

A leg fracture is a broken bone in the leg, which can occur due to various reasons such as falling or twisting the leg during an activity like hiking. Here are some necessary precautions and treatment steps for a person who has fractured their leg:

1. Assess the severity of the injury: If you suspect a leg fracture, it's essential to assess the extent of the injury. Check for signs of swelling, bruising, deformity, or inability to move the leg. If there is significant pain, instability, or open wounds, seek medical attention immediately.
2. Immobilize the affected limb: Use a splint, sling, or brace to immobilize the fractured leg to prevent further damage and promote healing. Be sure not to force the bone back into place as this could cause more harm.
3. Control bleeding: Apply direct pressure with a clean cloth to control any bleeding if present. Elevate the injured leg above heart level to reduce swelling and pain.
4. Pain management: Over-the-counter pain relievers like acetaminophen or ibuprofen can help manage pain. Prescription pain medication may be necessary in some cases, depending on the severity of the fracture.
5. Follow RICE protocol: Rest, Ice, Compression, and Elevation (RICE) are essential steps for managing a leg fracture. Rest the injured limb to prevent further damage, apply ice to reduce swelling, use compression bandages or splints to support the injury, and elevate the leg to minimize pain and promote healing.
6. Seek medical attention: It's crucial to consult a healthcare professional for proper diagnosis and treatment. X-rays will be necessary to confirm the fracture location and extent. Depending on the severity of the fracture, your doctor may recommend immobilization with a cast or brace, surgery, or other treatments.
7. Rehabilitation: After the initial injury has healed, rehabilitation exercises are essential for restoring strength, flexibility, and mobility to the affected leg. Your healthcare provider can recommend specific exercises and physical therapy sessions to help you recover fully.
8. Follow-up appointments: Regular follow-up appointments with your healthcare professional will be necessary to monitor progress, adjust treatment plans if needed, and ensure a complete recovery.
9. Preventing future injuries: Once

Data Preparation for RAG

```
#Libraries for processing dataframes, text
import json, os
import tiktoken
import pandas as pd
```

```
#Libraries for Loding Data, Chunking, Embedding, and Vector Databases
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings.sentence_transformer import
SentenceTransformerEmbeddings
from langchain_community.vectorstores import Chroma
```

Loading the Data

```
from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).

pdf_loader =
PyMuPDFLoader("/content/drive/MyDrive/AIML/medical_diagnosis_manual.pdf")

manual = pdf_loader.load()
```

Data Overview

Checking the first 5 pages

```
for i in range(5):
    print(f"Page {i+1}:", end='\n')
    print(manual[i].page_content, end='\n')
```

Page 1:
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Checking the number of pages

```
print(f"Number of pages: {len(manual)}")
```

Number of pages: 4114

Data Chunking

```
text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(
    encoding_name="cl100k_base",
    chunk_size=1024,
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document_chunks = pdf_loader.load_and_split(text_splitter)

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```

Embedding

```
embedding_model =
SentenceTransformerEmbeddings(model_name='thenlper/gte-large')

/tmp/ipykernel_18574/4198310515.py:1: LangChainDeprecationWarning: The
class `HuggingFaceEmbeddings` was deprecated in LangChain 0.2.2 and
will be removed in 1.0. An updated version of the class exists in
the :class:`~langchain-huggingface` package and should be used instead.
To use it run `pip install -U :class:`~langchain-huggingface` and
import as `from :class:`~langchain_huggingface import
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```

```
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embedding_model.embed_query(document_chunks[1].page_content)
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```

```
print("Dimension of embedding vector ", len(embedding_1))
len(embedding_1) == len(embedding_2)
```

Dimension of embedding vector 1024

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embedding_1, embedding_2

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```

Vector Database

```
output_dir = "/content/drive/MyDrive/AI/ML/vector_store"
```

```
if not os.path.exists(output_dir):
    os.makedirs(output_dir)
```

```
vector_store = Chroma.from_documents(
    document_chunks,
    embedding_model,
    persist_directory=output_dir
)
```

```
vector_store = Chroma(
    persist_directory=output_dir,
    embedding_function=embedding_model
)
```

```
/tmp/ipykernel_18574/1354474335.py:1: LangChainDeprecationWarning: The
class `Chroma` was deprecated in LangChain 0.2.9 and will be removed
in 1.0. An updated version of the class exists in
```

```

the :class:`~langchain-chroma` package and should be used instead. To
use it run `pip install -U :class:`~langchain-chroma` and import as
`from :class:`~langchain_chroma import Chroma`.
vector_store = Chroma(

vector_store.embeddings

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Chapter 222. Approach to the Critically Ill Patient

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model_output = llm(
    "What is the protocol for managing sepsis in a critical care unit?",
    max_tokens=512,
    temperature=0,
    echo=False
)

Llama.generate: prefix-match hit

model_output['choices'][0]['text'].strip()

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System and User Prompt Template

```
# Define the system prompt for the model
qna_system_message = """
You are an AI assistant designed to support professional doctors at
St. Bernard's Medical Center. Your task is to provide evidence-based,
concise, and relevant medical guidelines and best practices.

User input will include the necessary context for you to answer their
questions. This context will begin with the token: ###Context. The
context contains references to

When crafting your response:
1. Use only the provided context to answer the question.
2. If the answer is found in the context, respond with concise and
```

actionable medical insights.

3. Include the source reference with the page number, journal name, or publication, as provided in the context.

4. If the question is unrelated to the context or the context is empty, clearly respond with: "Sorry, this is out of my knowledge base."

Please adhere to the following response guidelines:

- Provide clear, direct answers using only the given context.
- Do not include any additional information outside of the context.

Avoid rephrasing or summarizing the context unless explicitly relevant to the question.

- If no relevant answer exists in the context, respond with: "Sorry, this is out of my knowledge base."

- If the context is not provided, your response should also be: "Sorry, this is out of my knowledge base."

Here is an example of how to structure your response:

Answer:

[Medical answer based on context]

Source:

[Source details with page or section]

"""

```
qna_user_message_template = """
```

```
### Context
```

```
Here are some excerpts from the medical manual:
```

```
{context}
```

```
### Question
```

```
{question}
```

```
"""
```

Response Function

```
def
generate_rag_response(user_input,k=1,max_tokens=128,temperature=0,top_
p=0.95,top_k=50):
    global qna_system_message,qna_user_message_template, vector_store
    # Create a temporary retriever with the desired k
    current_retriever =
vector_store.as_retriever(search_type="similarity",
search_kwargs={"k": k})
    relevant_document_chunks = current_retriever.invoke(user_input)
    context_list = [d.page_content for d in relevant_document_chunks]

    # Combine document chunks into a single context
```



```

context_for_query = ". ".join(context_list)

user_message = qna_user_message_template.replace('{context}',
context_for_query)
user_message = user_message.replace('{question}', user_input)

prompt = qna_system_message + '\n' + user_message

# Generate the response
try:
    response = llm(
        prompt=prompt,
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p,
        top_k=top_k
    )

    # Extract and print the model's response
    response = response['choices'][0]['text'].strip()
except Exception as e:
    response = f'Sorry, I encountered the following error: \n {e}'

return response

```

Question Answering using RAG

Query 1: What is the protocol for managing sepsis in a critical care unit?

```

user_input_1 = "What is the protocol for managing sepsis in a critical
care unit?"
print(generate_rag_response(user_input_1))

```

Llama.generate: prefix-match hit

Answer:

The context does not provide specific information on the protocol for managing sepsis in a critical care unit. However, it mentions that prevention of infection is an important aspect of supportive care for ICU patients (p. 3480). Early detection and treatment of sepsis are crucial to improve patient outcomes.

Source:

The context does not provide a specific source for information on the protocol for managing sepsis in a critical care unit.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
user_input_2 = "What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?"  
print(generate_rag_response(user_input_2))
```

Llama.generate: prefix-match hit

Answer:

The common symptoms for appendicitis include abdominal pain, usually localized in the right lower quadrant, loss of appetite, nausea, vomiting, constipation or diarrhea, and fever. Appendicitis cannot be cured via medicine alone; it requires surgical removal of the appendix (appendectomy).

Source:

The Merck Manual of Diagnosis & Therapy, 19th Edition, Chapter 11. Acute Abdomen & Surgical Gastroenterology, pages 165-167.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
user_input_3 = "What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?"  
print(generate_rag_response(user_input_3))
```

Llama.generate: prefix-match hit

Answer:

The treatment options for sudden patchy hair loss, also known as alopecia areata, include topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA) [Context: Table 86-2. Interpreting Findings in Alopecia]. Hormonal modulators such as oral contraceptives or spironol

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
user_input_4 = "What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or
```

```
permanent impairment of brain function?"  
print(generate_rag_response(user_input_4))
```

Llama.generate: prefix-match hit

Answer:

The initial treatment for traumatic brain injury (TBI) includes ensuring a reliable airway and maintaining adequate ventilation, oxygenation, and blood pressure. Surgery may be needed for patients with more severe injuries to place monitors to track and treat intracranial pressure, decompress the brain if intracranial pressure is increased, or remove intracranial hematomas (Chapter 324, The Merck Manual of Diagnosis & Therapy, p. 3395). In the first few days after the injury, maintaining adequate brain perfusion and oxygen

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
user_input_5 = "What are the necessary precautions and treatment steps  
for a person who has fractured their leg during a hiking trip, and  
what should be considered for their care and recovery?"  
print(generate_rag_response(user_input_5))
```

Llama.generate: prefix-match hit

Answer:

For a person with a fractured leg from a hiking accident, it's crucial to immobilize the affected limb using a splint or cast. This will help prevent further damage and allow the bone to heal properly. The patient should also apply ice to reduce swelling and pain. Seek medical care if an odor emanates from within the cast, a fever develops, or there is suspicion of infection (Merck Manual, 3382). Prolonged immobilization can lead to complications such as stiffness, contractures, and muscle atrophy;

Fine-tuning

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
fn_user_input_1 = "What is the protocol for managing sepsis in a  
critical care unit?"  
print(generate_rag_response(fn_user_input_1, temperature=0.5))
```

Llama.generate: prefix-match hit

Answer:

The context does not provide specific information on managing sepsis

in a critical care unit. However, it mentions that preventing infection is an essential part of supportive care for ICU patients (see p. 3480). Early detection and prompt treatment of sepsis are crucial to improve patient outcomes. Standard sepsis management includes administering appropriate antibiotics based on culture results, fluid resuscitation, and addressing underlying conditions.

Source:

Not applicable in this case as the context does not provide a specific reference for managing sepsis in the ICU.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
fn_user_input_2 = "What are the common symptoms for appendicitis, and  
can it be cured via medicine? If not, what surgical procedure should  
be followed to treat it?"  
print(generate_rag_response(fn_user_input_2,top_p=0.8,top_k=20))
```

Llama.generate: prefix-match hit

Answer:

The common symptoms for appendicitis include abdominal pain, usually localized in the right lower quadrant, loss of appetite, nausea, vomiting, constipation or diarrhea, and fever. Appendicitis cannot be cured via medicine alone; it requires surgical removal of the appendix (appendectomy).

Source:

The Merck Manual of Diagnosis & Therapy, 19th Edition, Chapter 11. Acute Abdomen & Surgical Gastroenterology, pages 165-167.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
fn_user_input_3 = "What are the effective treatments or solutions for  
addressing sudden patchy hair loss, commonly seen as localized bald  
spots on the scalp, and what could be the possible causes behind it?"  
print(generate_rag_response(fn_user_input_3,max_tokens=200))
```

Llama.generate: prefix-match hit

Answer:

The treatment options for sudden patchy hair loss, also known as alopecia areata, include topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or

psoralen plus ultraviolet A (PUVA) [Context: Table 86-2. Interpreting Findings in Alopecia]. Hormonal modulators such as oral contraceptives or spironolactone may be useful for female-pattern hair loss associated with hyperandrogenemia [Context: Table 86-2. Interpreting Findings in Alopecia]. If the underlying disorder is identified, it should be treated accordingly [Context: Hair loss due to other causes].

Source:
The Merck Manual of

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
fn_user_input_4 = "What treatments are recommended for a person who  
has sustained a physical injury to brain tissue, resulting in  
temporary or permanent impairment of brain function?"  
print(generate_rag_response(fn_user_input_4,top_p=0.85,top_k=20,temper  
ature=0.9))
```

Llama.generate: prefix-match hit

Answer:

The initial treatment for traumatic brain injury (TBI) involves ensuring a reliable airway, maintaining adequate ventilation, oxygenation, and blood pressure. Surgery may be needed in patients with more severe injuries to place monitors to track and treat intracranial pressure, decompress the brain if intracranial pressure is increased, or remove intracranial hematomas (Chapter 324, The Merck Manual of Diagnosis & Therapy, p. 3395). In the first few days after the injury, maintaining adequate brain perfusion and oxygen

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
fn_user_input_5 = "What are the necessary precautions and treatment  
steps for a person who has fractured their leg during a hiking trip,  
and what should be considered for their care and recovery?"  
print(generate_rag_response(fn_user_input_5))
```

Llama.generate: prefix-match hit

Answer:

For a person with a fractured leg from a hiking accident, it's crucial to immobilize the affected limb using a splint or cast. This will help prevent further damage and allow the bone to heal properly. The patient should also apply ice to reduce swelling and pain. Seek

medical care if an odor emanates from within the cast, a fever develops, or there is suspicion of infection (Merck Manual, 3382). Prolonged immobilization can lead to complications such as stiffness, contractures, and muscle atrophy;

Output Evaluation

Let us now use the LLM-as-a-judge method to check the quality of the RAG system on two parameters - retrieval and generation. We illustrate this evaluation based on the answers generated to the question from the previous section.

- We are using the same Mistral model for evaluation, so basically here the llm is rating itself on how well he has performed in the task.

```
groundedness_rater_system_message = """
You are tasked with rating AI generated answers to questions posed by
users.
You will be presented a question, context used by the AI system to
generate the answer and an AI generated answer to the question.
In the input, the question will begin with ###Question, the context
will begin with ###Context while the AI generated answer will begin
with ###Answer.

Evaluation criteria:
The task is to judge the extent to which the metric is followed by the
answer.
1 - The metric is not followed at all
2 - The metric is followed only to a limited extent
3 - The metric is followed to a good extent
4 - The metric is followed mostly
5 - The metric is followed completely

Metric:
The answer should be derived only from the information presented in
the context

Instructions:
1. First write down the steps that are needed to evaluate the answer
as per the metric.
2. Give a step-by-step explanation if the answer adheres to the metric
considering the question and context as the input.
3. Next, evaluate the extent to which the metric is followed.
4. Use the previous information to rate the answer using the evaluation
criteria and assign a score.

"""

relevance_rater_system_message = """
Task: Rate relevance of an AI-generated answer
```

```

1. Identify key aspects of the question
2. Check if the answer addresses these using the context
3. Verify all important points are included and no irrelevant info is added
4. Briefly explain alignment, missing, or extra details
5. Assign score (1-5):
1 = Not followed, 5 = Fully followed
"""

user_message_template = """
###Question
{question}

###Context
{context}

###Answer
{answer}
"""

def generate_ground_relevance_response(user_input, k=1,
max_tokens=128, temperature=0, top_p=0.95, top_k=50):
    global qna_system_message, qna_user_message_template

    # Step 1: Retrieve relevant chunks and truncate
    relevant_document_chunks =
retriever.get_relevant_documents(query=user_input, k=k)
    context_list = [d.page_content[:800] for d in
relevant_document_chunks]
    context_for_query = "\n\n".join(context_list)
    if not context_list:
        return "Sorry, no relevant context found.", "Sorry, no
relevant context found."

    # Step 2: Prepare user message
    user_message =
qna_user_message_template.format(context=context_for_query,
question=user_input)

    # Step 3: Generate answer
    prompt = f"[INST]{qna_system_message}\nuser:
{user_message}[/INST]"
    response = llm(prompt=prompt, max_tokens=max_tokens,
temperature=temperature,
                    top_p=top_p, top_k=top_k, stop=['INST'])
    answer = response["choices"][0]["text"][:500] # truncate to avoid
overflow

    # Step 4: Prepare groundedness and relevance prompts
    groundedness_prompt = f"[INST]{groundedness_rater_system_message}\

```

```

nuser: {user_message_template.format(context=context_for_query,
question=user_input, answer=answer)}/{/INST}"
    relevance_prompt = f"[INST]{relevance_rater_system_message}\nuser:
{user_message_template.format(context=context_for_query,
question=user_input, answer=answer)}/{/INST}"

    # Step 5: Get ratings
    response_1 = llm(prompt=groundedness_prompt,
max_tokens=max_tokens, temperature=temperature,
                    top_p=top_p, top_k=top_k, stop=['INST'])
    response_2 = llm(prompt=relevance_prompt, max_tokens=max_tokens,
temperature=temperature,
                    top_p=top_p, top_k=top_k, stop=['INST'])

    return response_1['choices'][0]['text'], response_2['choices'][0]
['text']

```

Query 1: What is the protocol for managing sepsis in a critical care unit?

```

ground, rel = generate_ground_relevance_response(user_input_1,
max_tokens=350)
print(ground, end="\n\n")
print(rel)

```

```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit

```

Steps for evaluation:

1. Identify if the answer is derived only from the information presented in the context.
2. Check if the context provides any specific information related to the protocol for managing sepsis in a critical care unit.
3. Determine if the AI generated answer mentions anything outside of the provided context.

Explanation:

The context does not provide any specific information on the protocol for managing sepsis in a critical care unit. The AI generated answer correctly identifies this and does not mention anything outside of the provided context.

Evaluation:

The metric is not followed at all as there is no information in the context about the protocol for managing sepsis, and the AI generated answer does not provide any such information either.

Rating:

1 - The metric is not followed at all.

###Answer Evaluation:

1. Identify key aspects of the question: The question asks about the protocol for managing sepsis in a critical care unit.
2. Check if the answer addresses these using the context: The answer acknowledges that the context does not provide specific information on the protocol for managing sepsis in a critical care unit.
3. Verify all important points are included and no irrelevant info is added: N/A (The question was about the protocol, and the answer acknowledged that there was no relevant information provided in the context.)
4. Briefly explain alignment, missing, or extra details: The answer is aligned with the question as it acknowledges that the context does not provide the required information. However, it misses providing any information on the sepsis protocol itself.
5. Assign score (1-5): 1. The answer did not follow the question's requirement of providing information about the sepsis protocol in a critical care unit.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
ground, rel = generate_ground_relevance_response(user_input_2,
max_tokens=350)
print(ground, end="\n\n")
print(rel)
```

```
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
```

Steps to evaluate the answer:

1. Identify the key symptoms mentioned in the context for appendicitis.
2. Determine if the AI generated answer includes those symptoms.
3. Check if the AI generated answer mentions that appendicitis cannot be cured via medicine and requires surgical removal.
4. Verify if the surgical procedures mentioned in the answer (open or laparoscopic appendectomy) are consistent with the context.
5. Ensure that the information in the answer is derived only from the context.

Explanation:

The AI generated answer adheres to the metric as it includes the common symptoms for appendicitis mentioned in the context (abdominal pain, usually localized in the right lower quadrant, anorexia, nausea, and vomiting) and correctly states that appendicitis cannot be cured

via medicine alone and requires surgical removal. The answer also mentions the specific surgical procedures for treating appendicitis (open or laparoscopic appendectomy), which are consistent with the context.

Rating:

Based on the evaluation criteria, I would rate the answer as a 5 because it follows the metric completely by deriving all the information from the context.

###Rating

1. Identify key aspects of the question: The question asks about the common symptoms of appendicitis and whether it can be cured via medicine or if surgery is required.
2. Check if the answer addresses these using the context: Yes, the answer correctly identifies the common symptoms of appendicitis and states that surgery is necessary for treatment.
3. Verify all important points are included and no irrelevant info is added: The answer includes all the important points from the question and does not add any irrelevant information.
4. Briefly explain alignment, missing, or extra details: The answer aligns perfectly with the context provided as it correctly identifies the symptoms of appendicitis and states that surgery is required for treatment. No missing or extra details are present.
5. Assign score (1–5): 5. The answer fully follows the instructions provided in the question and accurately reflects the information given in the context.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
ground, rel = generate_ground_relevance_response(user_input_3,
max_tokens=350)
print(ground, end="\n\n")
print(rel)
```

```
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
```

Steps to evaluate the answer:

1. Identify the key information about effective treatments for sudden patchy hair loss from the context.
2. Determine if the AI generated answer is derived only from the information presented in the context.
3. Check if the possible causes of sudden patchy hair loss are mentioned in the context and if they are included in the answer.

Evaluation:

The AI generated answer mentions Finasteride as an effective treatment for certain types of hair loss, which is consistent with the information presented in the context. However, it does not explicitly state that Finasteride is a treatment for sudden patchy hair loss or localized bald spots. Additionally, the possible causes of sudden patchy hair loss are not mentioned in the answer.

Rating:

Based on the evaluation criteria and metric provided, I would rate the answer as follows:

3 - The metric is followed to a good extent. While the answer does not explicitly state that Finasteride is a treatment for sudden patchy hair loss or localized bald spots, it is mentioned in the context as an effective treatment for hair loss, which is the main focus of the question. However, the possible causes are not included in the answer.

be one of the causes as mentioned in the context. Therefore, the answer is relevant as it correctly identifies Finasteride as an effective treatment for certain types of hair loss (specifically androgenetic alopecia) and explains its mechanism of action. However, it does not fully address the question as it does not mention any other treatments or solutions for addressing sudden patchy hair loss besides Finasteride, nor does it discuss possible causes beyond androgenetic alopecia. Therefore, I would assign a score of 3 out of 5. The answer is partially relevant but could be improved by including additional information about other potential causes and treatments for sudden patchy hair loss.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
ground, rel = generate_ground_relevance_response(user_input_4,
max_tokens=350)
print(ground, end="\n\n")
print(rel)
```

```
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
```

Steps to evaluate the answer:

1. Identify the key information in the context related to treatments for traumatic brain injury (TBI).
2. Determine if the AI generated answer includes only the information from the context.
3. Evaluate the extent to which the answer adheres to the metric.

Explanation:

The context provides information about the initial treatment for TBI, including ensuring a reliable airway and maintaining adequate ventilation, oxygenation, and blood pressure. Surgery may also be needed to treat intracranial pressure or remove hematomas. In the first few days after the injury, maintaining brain perfusion and oxygenation is important.

The AI generated answer includes all of the information from the context related to initial treatments for TBI. Therefore, the answer adheres to the metric as it is derived only from the information presented in the context.

Rating:

Based on the evaluation criteria, I would rate this answer as a 5 because the metric is followed completely. The AI generated answer includes all of the relevant information from the context and does not include any additional or irrelevant information.

###Rating

1. Identified key aspects: The question asks about treatments for a person with traumatic brain injury (TBI), so the key aspects are identifying the specific injuries and the recommended treatments.
2. Relevance to question: The answer mentions the initial treatment for TBI, which includes ensuring a reliable airway, maintaining adequate ventilation, oxygenation, and blood pressure, as well as potential surgery to address intracranial pressure or hematomas. These are all relevant treatments mentioned in the context.
3. Completeness: The answer covers the initial treatment for TBI, which is in line with what's stated in the context. It does not mention any irrelevant information.
4. Alignment: The answer aligns well with the question and context as it discusses the treatments for a person with traumatic brain injury.
5. Missing details: None identified.
6. Extra details: None identified.
7. Score: 5 (Fully followed)

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
ground, rel = generate_ground_relevance_response(user_input_5,
max_tokens=350)
print(ground, end="\n\n")
print(rel)
```

```
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit  
Llama.generate: prefix-match hit
```

Actionable Insights and Business Recommendations

1. **Improve Contextual Relevance:** Increase the `chunk_overlap` setting in the retriever to enhance the relevance of retrieved results. Because medical manuals often follow step-by-step sequences, a larger overlap helps preserve continuity and context between chunks.
2. **Sustain Strong Groundedness:** The model currently performs well in groundedness due to carefully designed prompts, so maintaining this structured prompting approach is essential.
3. **Enhance Embedding Performance for Medical Use:** Although the existing embedding model is effective, adopting one trained specifically on medical data can further boost retrieval accuracy and domain relevance.
4. **Keep Knowledge Up to Date:** Regular updates to the knowledge base are important to incorporate the latest medical research, protocols, and clinical guidelines, ensuring ongoing accuracy.
5. **Add Multilingual Capabilities:** Introduce support for multiple languages to better serve healthcare professionals across different regions and linguistic backgrounds.
6. **Integrate User Feedback:** Build a feedback system that allows doctors to review and refine responses, helping the chatbot improve based on real-world usage and clinical needs.
7. **Extend to More Medical Fields:** Scale the RAG system to cover additional medical specialties, increasing its usefulness across a wider range of healthcare applications.

Power Ahead ____